Parameter adjustment

Experimental basis

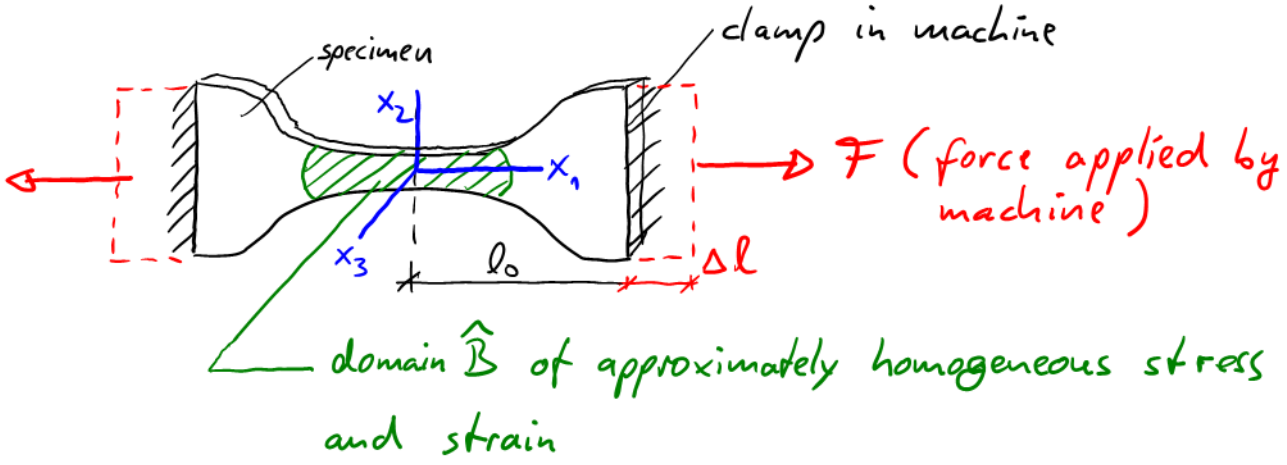
⇝ data regarding mechanical response at <u>material point</u>

Problem: Measurement at single material points (with infinitesimal volume) is difficult!

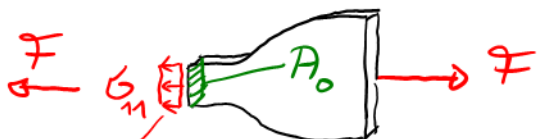
Solution: Construction of experiments, which lead to <u>homogeneous</u> fields of stress and strain at least in domains of measurable size.

Exple 1: Uniaxial tension

Dogbone specimen:



1) Identify stress in $\hat{\mathcal{B}}$ from equilibrium of forces:



homogeneous stress distribution in $\hat{\mathcal{B}}$

$A_0 \triangleq$ cross-section area in undeformed configuration

- physical stress $\sigma = \frac{F}{A}$ (force per deformed area)
- technical stress $\sigma = \frac{F}{A_0}$ (force per undeformed area)

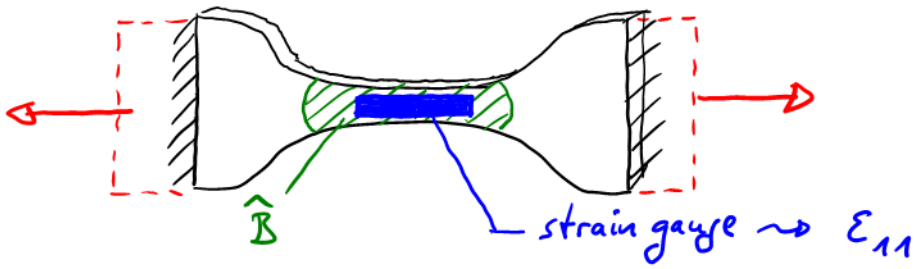
here: both are equal because small deformations are assumed ($A = A_0$)

$$\Rightarrow \sigma_{11} = \frac{F}{A_0}, \quad \sigma_{22} = \sigma_{33} = \sigma_{12} = \sigma_{13} = \sigma_{23} = 0$$

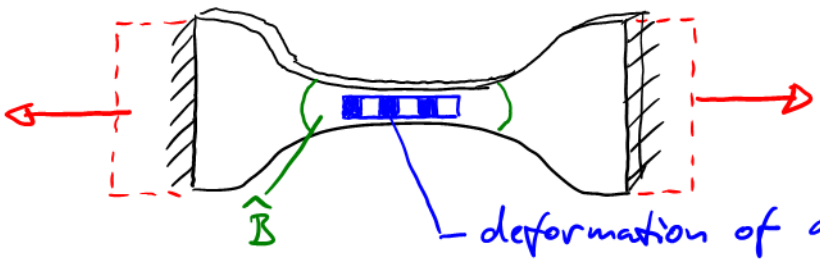
2) Identify strains by measurement in $\hat{\mathcal{B}}$

Attention: $\varepsilon_{11} \neq \frac{\Delta l}{l_0}$!!! (inhomogeneous distribution of $\underline{\varepsilon}$ outside $\hat{\mathcal{B}}$, this would only be average, not material point response)

- Measurement using strain gauge in $\hat{\mathcal{B}}$:



- Optical measurement using camera:



Alternative: Digital Image Correlation (DIC)

$\rightsquigarrow \varepsilon_{11}$ /

$\varepsilon_{12} = \varepsilon_{13} = \varepsilon_{23} = 0$

ε_{22} and ε_{33} are often not measured

$\Rightarrow$ Experimental data basis:

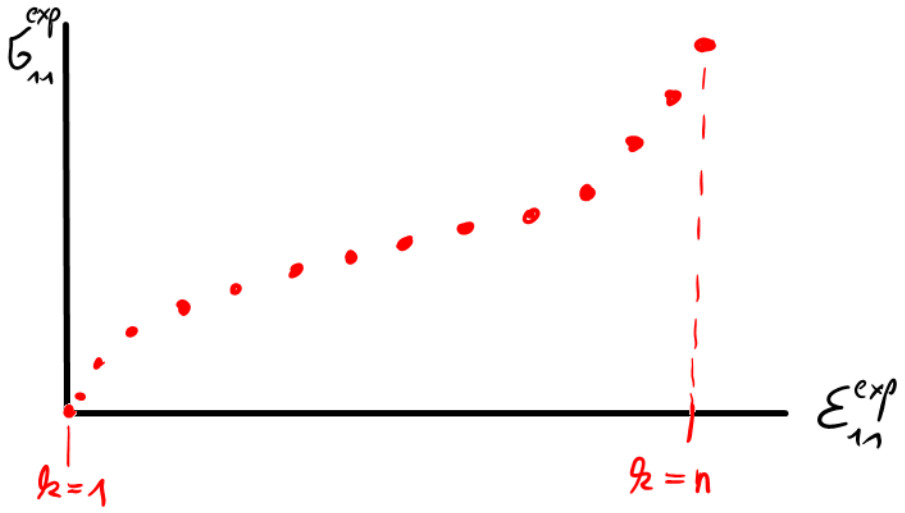
σ_{11}^{exp} ; (prescribed force)

$\sigma_{22}^{exp} = \sigma_{33}^{exp} = \sigma_{12}^{exp} = \sigma_{13}^{exp} = \sigma_{23}^{exp} = 0$

ε_{11}^{exp} ; (measured)

$\varepsilon_{12}^{exp} = \varepsilon_{13}^{exp} = \varepsilon_{23}^{exp} = 0$

for n-many measurement values $k = 1, 2, \dots n$



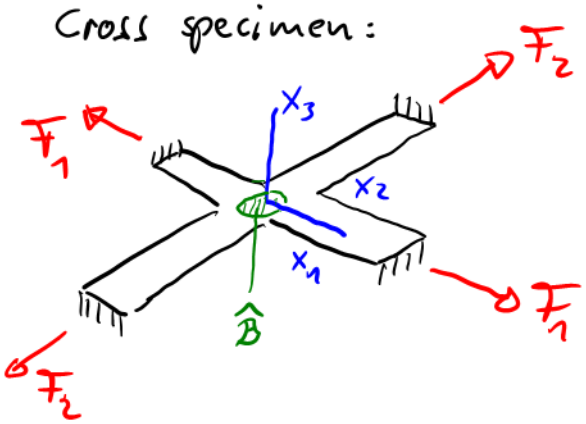
$\rightsquigarrow$ Poisson ratio can only be determined, if ε_{22} and ε_{33} are measured.

$\rightsquigarrow$ More information is obtained if tests are performed in different directions $\rightsquigarrow$ anisotropic properties.

$\rightsquigarrow$ Interaction response between different directions only quantifiable if lateral contractions are measured.

Example 2: Biaxial tension

Cross specimen:



Experimental basis in $\hat{B}$:

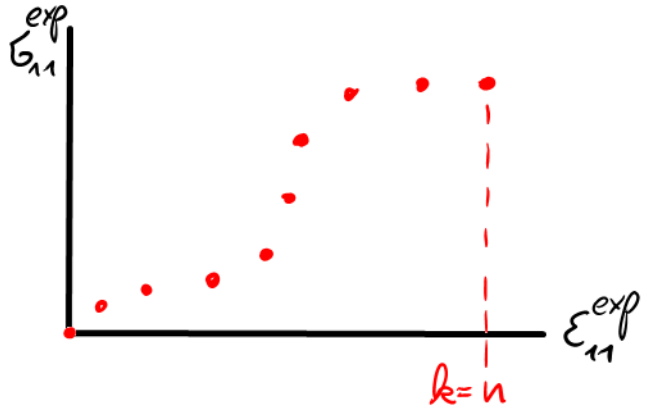
- $\sigma_{11}^{exp}, \sigma_{22}^{exp}$ (applied force)

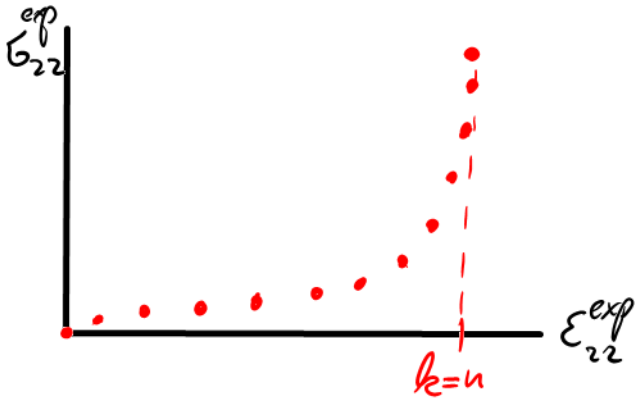
$$\sigma_{33}^{exp} = \sigma_{12}^{exp} = \sigma_{13}^{exp} = \sigma_{23}^{exp} = 0$$

- $\varepsilon_{11}^{exp}, \varepsilon_{22}^{exp}$ (measured)

$$\varepsilon_{12}^{exp} = \varepsilon_{13}^{exp} = \varepsilon_{23}^{exp} = 0$$

unknown: ε_{33}^{exp}





↝ contains information regarding interaction of directions x_1 and x_2

Parameter adjustment

By minimization of objective function ⇝ least square errors

- Objective / error function:

$$e(\underline{\underline{\alpha}}) := \frac{1}{\max_k \left[\sigma_k^{exp}\right]^2} \sum_k \left(\sigma_k^{mod}(\underline{\underline{\varepsilon}}^{exp}, \underline{\underline{\alpha}}) - \sigma_k^{exp} \right)^2 \longrightarrow \min_{\underline{\underline{\alpha}}}$$

- set of material parameters $\underline{\underline{\alpha}} = \left[\alpha_1, \alpha_2 \dots \alpha_{n_{para}}\right]^T$
- stress quantity of interest σ (e.g. uniaxial tension: σ_{11})

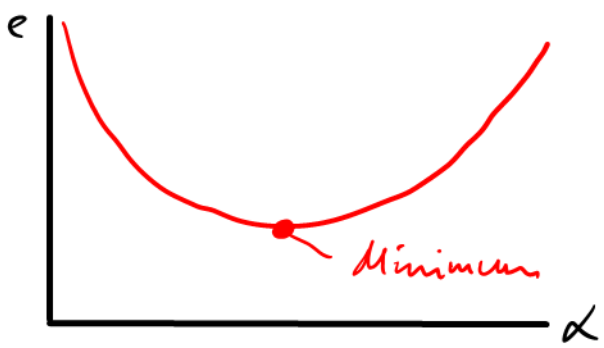
- Mostly, n_{exp}-many experiments are needed to identify all parameters

$$\rightsquigarrow e(\underline{\underline{\alpha}}) = \sum_{m=1}^{n_{exp}} e_m(\underline{\underline{\alpha}}) \longrightarrow \min_{\underline{\underline{\alpha}}}$$

expl: 1 biax.- and 1 uniax. test:

$$e(\underline{\underline{\alpha}}) = e_1^{Biax}(\underline{\underline{\alpha}}) + e_2^{Uni}(\underline{\underline{\alpha}})$$

- If $\underline{\underline{\sigma}}(\underline{\underline{\varepsilon}})$ is linear function in $\underline{\underline{\alpha}}$, then e is quadratic in $\underline{\underline{\alpha}}$ ⇝ convex in $\underline{\underline{\alpha}}$



- unique minimum
  ⇝ local minimum is global minimum
- Gradient-based optimizer yields global minimum

Minimum can then be computed from

$$\frac{\partial e}{\partial \alpha_\rho} = 0 \quad \text{for} \quad \rho = 1 \dots n_{para}$$

linear in α_ρ for quadratic $e(\underline{\alpha})$

$\rightsquigarrow$ yields system of linear equations

Attention: If minimum is in non-feasible regime of values, e.g. negative value for Young's modulus ($E < 0$), then alternative optimization strategy for $e \rightarrow \min$ is needed which accounts for prescribed bounds, e.g. $E \overset{!}{>} 0$.
$\rightsquigarrow$ e.g. "quadratic programming"

Examples:

a) Linear elasticity:

$$\underline{\sigma} = \lambda \, \mathrm{tr}\,\underline{\varepsilon}\, \underline{1} + 2\mu\, \underline{\varepsilon}$$

linear in λ, μ ✓

b) Nonlinear elasticity:

$$\underline{\sigma} = \lambda \, \mathrm{tr}\,\underline{\varepsilon}\, \underline{1} + 2\mu\, \underline{\varepsilon} + \varkappa_{fib}\, J_1\, \underline{1} + \mu_{fib}\, J_2^2\, \mathrm{dev}\,\underline{\varepsilon}$$

linear in $\lambda, \mu, \varkappa_{fib}, \mu_{fib}$ ✓

(despite nonlinearity in $\underline{\varepsilon}$)

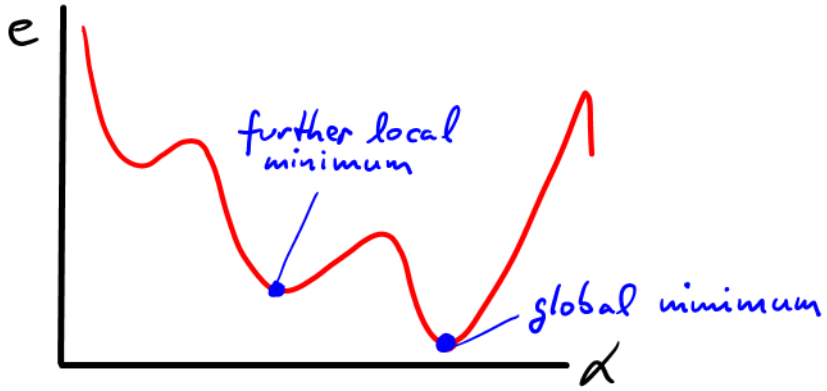
- If $\underline{\underline{\sigma}}(\underline{\underline{\varepsilon}})$ is nonlinear in $\underline{\underline{\alpha}}$:

  Expl.: $\underline{\underline{\sigma}} = \lambda \operatorname{tr}\underline{\underline{\varepsilon}}\,\underline{\underline{1}} + 2\mu\underline{\underline{\varepsilon}} + \varkappa_{fib}\underline{\underline{\mathcal{F}}}_1^{\alpha}$

  → linear in $\lambda, \mu, \varkappa_{fib}$

  → but nonlinear in α

⇒ In general, e can not be assumed to be convex in $\underline{\underline{\alpha}}$!



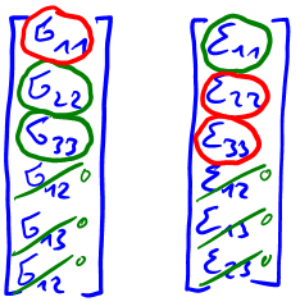
→ application of global (nonconvex) optimization methods

→ gradient-based minimizers do find local minima

→ finding global minimum is not guaranteed

Model response in line with selected experiment

- Calculation of e requires $\underline{\underline{\sigma}}^{mod}(\underline{\underline{\varepsilon}}^{exp})$, but not all components of $\underline{\underline{\varepsilon}}^{exp}$ may be known
  e.g. uniaxial tension: ε_{22}^{exp} and ε_{33}^{exp} are often not measured and thus unknown, however, σ_{11} does depend on ε_{22} and ε_{33}



○ ≙ known from experiment

○ ≙ to be computed

⇝ calculation of ε_{22} and ε_{33} from $\sigma_{22} \stackrel{!}{=} 0$ and $\sigma_{33} \stackrel{!}{=} 0$

- Analytically only possible in special cases, e.g. linear, isotropic elasticity in uniaxial tension

$$\underline{\underline{\sigma}} = \underline{\underline{\mathbb{C}}}\ \underline{\underline{\varepsilon}} \rightsquigarrow \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \hline \vdots \end{bmatrix} = \left[\begin{array}{ccc|c} \lambda+2\mu & \lambda & \lambda & \underline{\underline{0}} \\ \lambda & \lambda+2\mu & \lambda & \\ \lambda & \lambda & \lambda+2\mu & \\ \hline \multicolumn{3}{c|}{\underline{\underline{0}}} & \ddots \end{array}\right] \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \hline \vdots \end{bmatrix}$$

$$\rightsquigarrow \sigma_{22} = \lambda\,\varepsilon_{11} + (\lambda+2\mu)\,\varepsilon_{22} + \lambda\,\varepsilon_{33} \stackrel{!}{=} 0 \quad (1)$$

$$\sigma_{33} = \lambda\,\varepsilon_{11} + \lambda\,\varepsilon_{22} + (\lambda+2\mu)\,\varepsilon_{33} \stackrel{!}{=} 0 \quad (2)$$

$$(1): \quad \varepsilon_{33} = -\varepsilon_{11} - \frac{\lambda+2\mu}{\lambda}\,\varepsilon_{22} \quad (3)$$

(3) in (2): $\lambda \varepsilon_{11} + \lambda \varepsilon_{22} + (\lambda+2\mu)\left(-\varepsilon_{11} - \frac{\lambda+2\mu}{\lambda}\varepsilon_{22}\right) = 0$

$\Leftrightarrow \lambda \varepsilon_{11} + \lambda \varepsilon_{22} - \lambda \varepsilon_{11} - (\lambda+2\mu)\varepsilon_{22}$

$-2\mu \varepsilon_{11} - \frac{2\mu(\lambda+2\mu)}{\lambda}\varepsilon_{22} = 0$

$\Leftrightarrow \varepsilon_{11}\underbrace{(\lambda - \lambda - 2\mu)}_{-2\mu} + \varepsilon_{22}\underbrace{\left(\lambda - (\lambda+2\mu) - \frac{2\mu(\lambda+2\mu)}{\lambda}\right)}_{\frac{-2\mu\lambda - 2\mu\lambda - 4\mu^2}{\lambda} = -4\mu - 4\frac{\mu^2}{\lambda}} = 0$

$\Leftrightarrow \varepsilon_{22} = -2\mu\, \varepsilon_{11} \frac{1}{2(2\mu + 2\frac{\mu^2}{\lambda})} = -\frac{1}{2+2\frac{\mu}{\lambda}}\varepsilon_{11}$

in (3): $\varepsilon_{33} = -\varepsilon_{11} - \frac{\lambda+2\mu}{\lambda}\varepsilon_{22}$

$= -\varepsilon_{11} + \frac{1+2\frac{\mu}{\lambda}}{1}\frac{1}{2+2\frac{\mu}{\lambda}}\varepsilon_{11}$

$= \left(\frac{1+2\frac{\mu}{\lambda} - 2 - 2\frac{\mu}{\lambda}}{2+2\frac{\mu}{\lambda}}\right)\varepsilon_{11}$

$= -\frac{1}{2+2\frac{\mu}{\lambda}}\varepsilon_{11} = \varepsilon_{22}$ ✓ (isotropy)

- If analytic calculation is not possible, then numerical calculation e.g. with Newton iteration.